CHAPTER 1: Motor Control

Outline of Chapter 1:

1. Motor Fundamentals.
2. STSPIN240 Motor Driver
3. Pulse-Width-Modulation (PWM) using the STM32 MCU to control the speed of a motor.
4. STM32 Timer encoder mode to read the velocity of a motor.
5. Moving Average Filter.
6. Proportional-Integral-Differential (PID) Motor Control.

Motor Types

1. DC brushed motor - Robotic small-scale vehicles
2. Brushless DC motor - Drones
3. Stepper motor - CNC Machine, 3d Printers

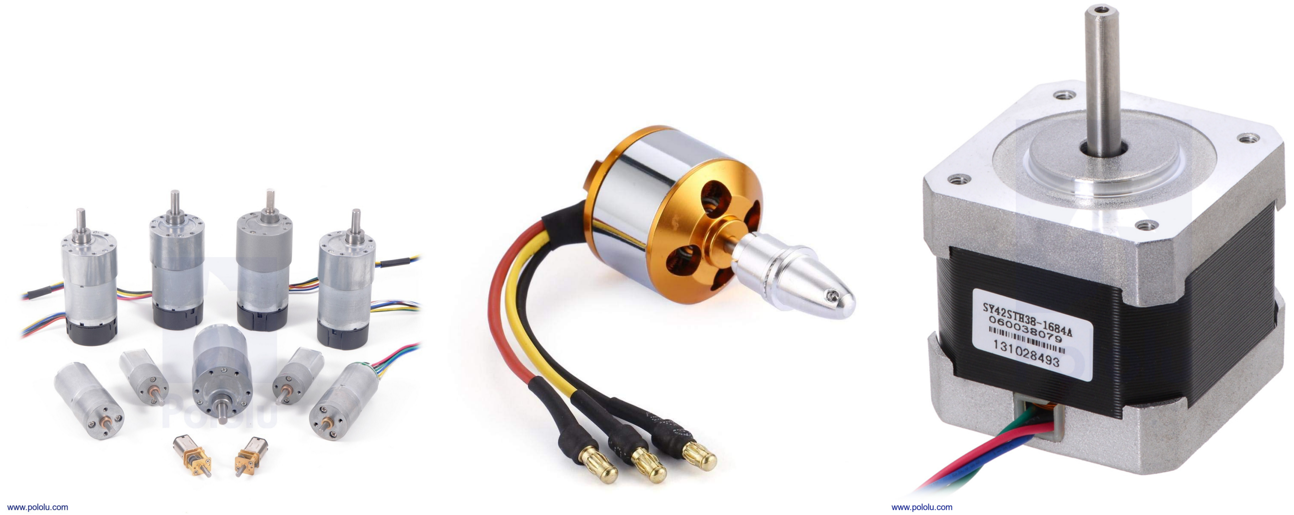


Figure: DC motor, Brushless DC Motor, Stepper Motor

Brushed DC motor Internal Structure

1. Magnet - stator
2. Coils with a shaft - rotor
3. Commutator and Brushes - ensure electrical connection

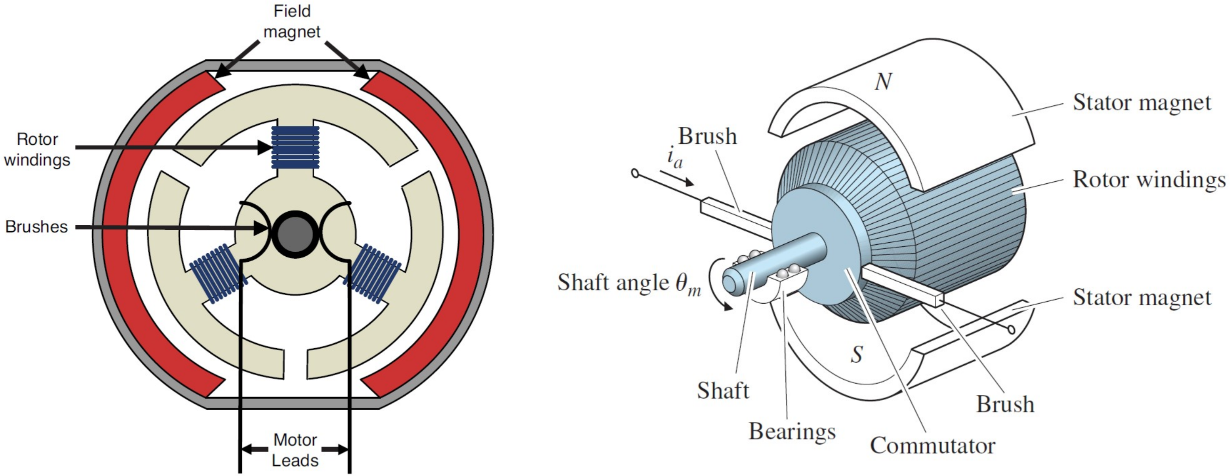


Figure: Internal Structure of the DC motor

Basic Principles: Torque and Rotational speed

Rotational Speed,angular velocity, defines how fast a motor is rotating. Its unit is rotation per minute (rpm, in practical cases) or radians/sec(SI unit).

Torque is a measure of the rotational force applied to an object, causing it to rotate around an axis. In the context of a motor, torque is the force that produces or resists rotation. Torque is equal to the product of the force(F) and the distance (r) from the axis of the rotation. Equation:

$$T = F \times r \times \sin(\theta), \qquad (1)$$

where T is torque, the lever arm length, θ is the angle between the force and the lever arm.

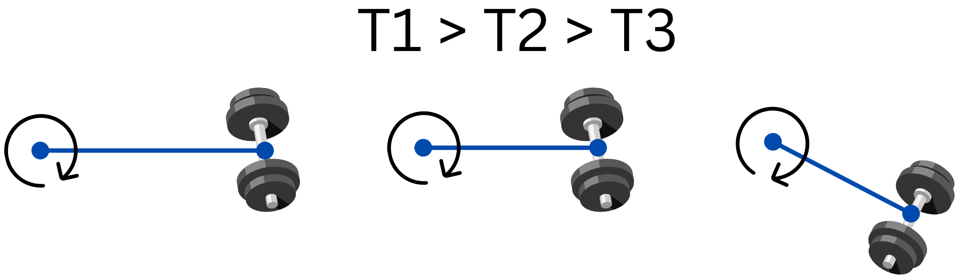


Figure: Torque illustration

Torque-speed curve

The Torque is inverse proportional to the speed.
No-load speed is the maximum speed that the motor can rotate with.
Stall torque is the maximum torque that a motor can produce when the rotor (armature) is prevented from rotating.

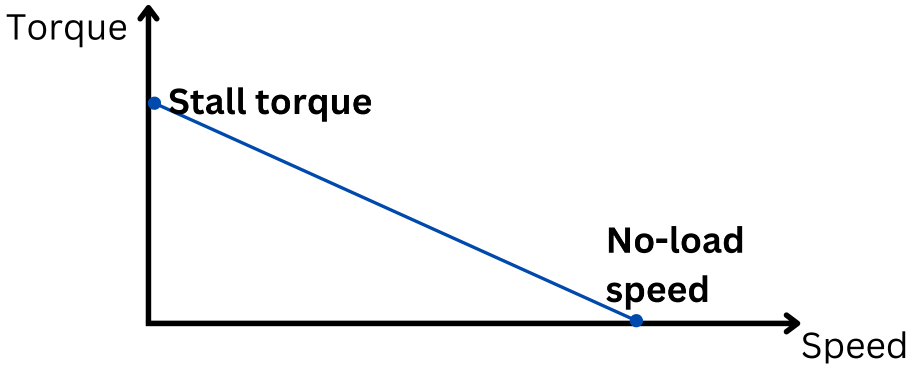


Figure: Torque-speed curve

Why do we need a gearbox?

Using the gearbox allows us to adjust torque-speed ratio to produce more torque. The gearbox ratio dictates

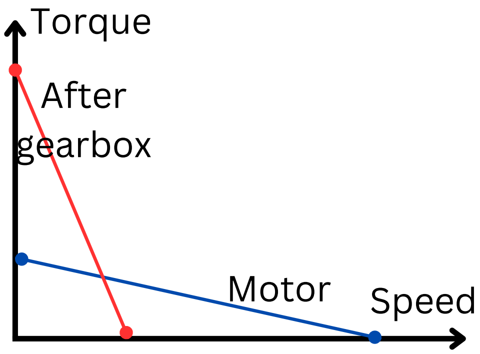


Figure: How gearbox chnages torque-speed curve

Torque-current and Speed-voltage relation

Torque is proportional to the applied current:

$$T = K_t i_a \tag{2}$$

Speed is proportional to the armature voltage:

$$e = K_e \dot{\theta}_m \tag{3}$$

Motor Speed:

$$\dot{\theta}_m = \frac{v_a - TR_a/K_t}{K_e} \tag{4}$$

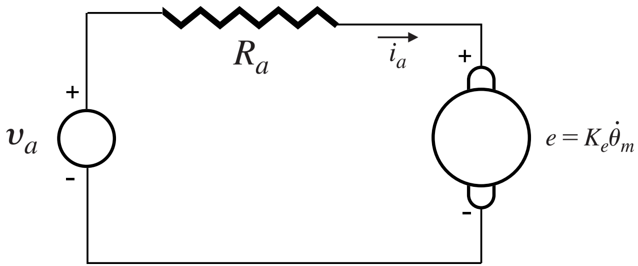


Figure: An electric circuit of an armature

MOSFET?

A transistor is necessary to provide enough current to drive the motor. The microcontroller cannot generate a usfficient current level to drive the motors.

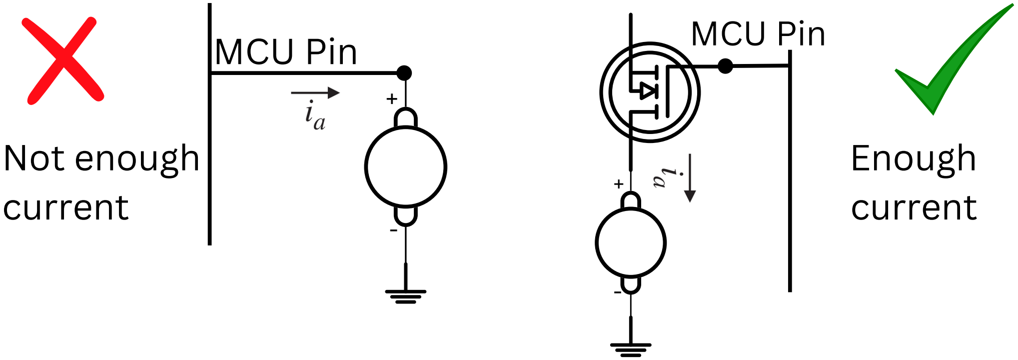


Figure: Direct connection vs MOSFET

H-bridge

H-bridge is necessary to control the direction of the motor rotation

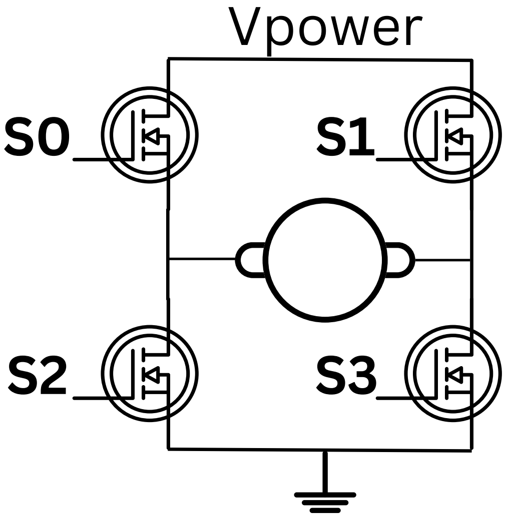
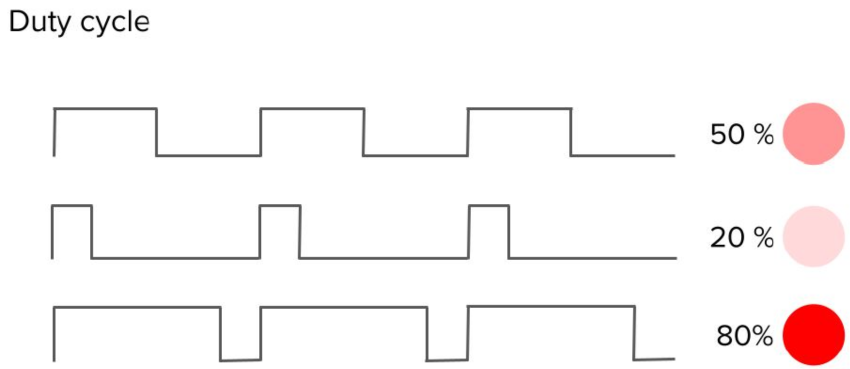


Figure: H-bridge connection

PWM signal

Pulse-width-modulation signal is needed to generate an analog signal using the digital GPIO pin.



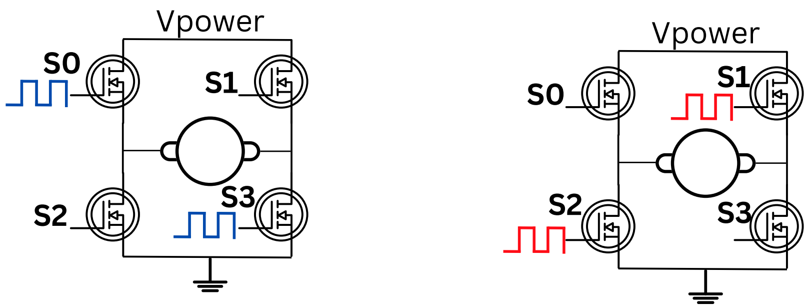


Figure: H-bridge connection

Motor Drivers

Motor Driver is an electronic circuit to provide a sufficient level current to driver a motor. It allows to control the motor using a low-power signal, including PWM. In addition, it ensures over-current and over-temerature protection. Motors Drivers:

1. STSPIN240 - dual channel, 1.3 A, STMicroElectronics
2. STSPIN250 - single channel, 2.5 A,
3. TB6612 - dual channel, 1.2 A, Sparkfun
4. TB6612 - dual channel, 1.2 A, Adafruit
5. MC33926 - single/dual channel, 3A, Pololu
6. L298N - dual channel, 2A, DFRobot